

Architecture-Controlled Ring-Opening Polymerization for Dynamic Covalent Poly(disulfide)s

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Supporting Information

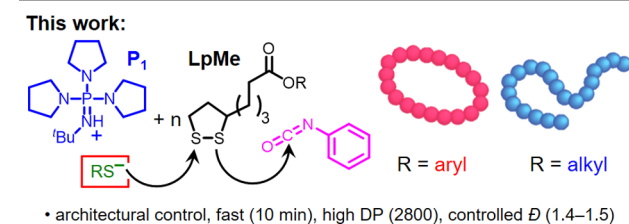
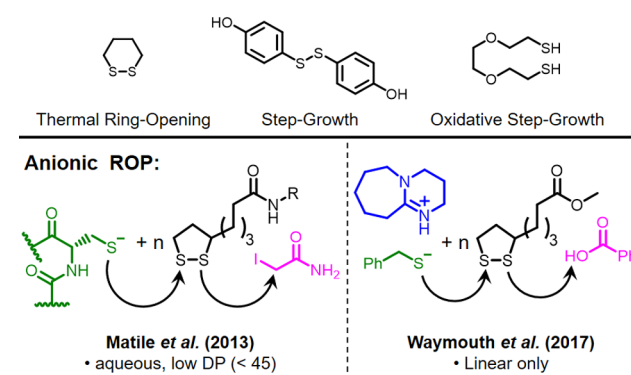
ABSTRACT: A strategy is reported for controlling the architecture of poly(disulfide)s by ring-opening polymerization. Aryl thiol initiators shift the ring–chain equilibrium to yield cyclic polymers, while alkyl thiols favor linear ones. Control over polymerization enables synthesis of large polymers (630 kDa) and catalytic depolymerization to recycle monomers. This work provides a new avenue to create dynamic covalent polymers with controlled geometry and length, allowing better characterization of structure–property relationships to expand their materials potentials.

Disulfide (S–S) bonds are widely present in biology, chemistry, and materials science. As a well-studied dynamic covalent system, disulfides undergo rapid and reversible exchange with disulfides or thiolates in a rather media-insensitive manner.¹ The ability to wield the reversible covalent chemistry via external stimuli makes the dynamic disulfide bond useful in a range of applications, such as controlling protein conformations,² mediating cellular redox homeostasis,^{3,4} cellular delivery of drugs,^{5–7} biomolecules,^{8–10} and liposomes,¹¹ dynamic self-assembly,^{12–14} self-healing materials,^{15,16} inorganic/organic hybrid polymers,^{17,18} reprocessable vitrimers,¹⁹ CO₂ regulation,²⁰ and Li–S batteries.²¹ This versatility of disulfide chemistry in materials relies on the synthesis of disulfide-containing polymers, i.e., poly(disulfide)s.²² Conventional methods include thermally triggered ring-opening polymerization,^{23,24} step-growth polymerization of disulfide-containing monomers,²⁵ or oxidative step-growth of dithiols^{26,27} (Scheme 1). The lack of controlled synthesis of poly(disulfide)s limits studies on structure–property relationship of disulfide-based functional materials.

Recently, Matile^{28,29} and Waymouth³⁰ demonstrated the synthesis of linear poly(disulfide)s using anionic ring-opening polymerization (ROP). However, methods remain absent for selective and controlled synthesis of different macromolecular architectures of poly(disulfide)s, e.g., cyclic topology. Cyclic polymers^{31,32} are generally obtained by catalyst-driven ring expansion,^{33–37} backbiting of reactive chain ends,^{38–45} or switching macromolecular interlock molecules.^{46,47} One key challenge for preparing cyclic poly(disulfide)s via ROP is unselective dynamic exchange between thiolate chain ends and

Scheme 1. Methods for the Synthesis of Poly(Disulfide)s

Conventional Methods:

^aDP = degree of polymerization, LpMe = methyl lipate.

disulfide backbone, i.e., chain transfer. Similar issue of chain transfer is faced by other sulfur-based ROP.⁴⁸

Here we address the challenge of controlling the poly(disulfide) architecture to selectively synthesize cyclic or linear chains and, thereby expand their materials potential. We wondered if ROP initiators that are also good leaving groups would generate cyclic polymers based on the hypothesis that a facile leaving group promotes ring formation early in the polymerization via backbiting and maintains such preference for the duration of the reaction. The propagation is presumed to proceed via catalytic scission of rings insofar as the jettisoned nucleofuge is capable of reinitiation. To test this hypothesis, we used phenylthiol (PhSH) as the initiator given its better nucleofugality compared to alkyl ones. (±)-Methyl

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1) Base / THF / r.t.
2) PhNCO

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initiator (I)	base (B)	conversion (%) ^b	M _n (kDa) ^c	D ^c	topology
PhSH	Me ₄ -guanidine	73	22	1.4	cyclic
PhSH	DBU	68	24	1.5	cyclic
PhSH	TBD	65	52	1.4	cyclic
PhSH	P ₄ - <i>t</i> -Bu	73	65	1.4	cyclic
PhSH	P ₁	72	38	1.4	cyclic
<i>p</i> -MeOPhSH	P ₁	71	29	1.5	cyclic
BnSH	P ₁	72	18	1.4	linear
C ₁₂ H ₂₅ SH	P ₁	66	15	1.5	linear
<i>p</i> -NO ₂ PhSH	P ₁	21	oligomers		
PhSH	pyridine	11	195	1.6	

A

BnSH / P₁

PhSH / P₁

Chemical Shift (ppm)

B

$n = 6$, $\Delta m = 220.3$

$n = 7$, $m = 220.3 \cdot n + 23$

$n = 8$, Monomer Na⁺

$n = 9$

$n = 10$

$n = 12$

$n = 16$

$n = 22$

m/z

C

$\log M_w (\text{g mol}^{-1})$

Retention time (min)

D

Weight (%)

Temperature (°C)

Mass Spec Signal @ $m/z = 220$

Consistent with other ROPs,⁵² synthesis of poly(LpMe) is favored at higher concentrations (Figure 2A) and lower temperatures (Figure 2B). The conversion for cyclic polymer

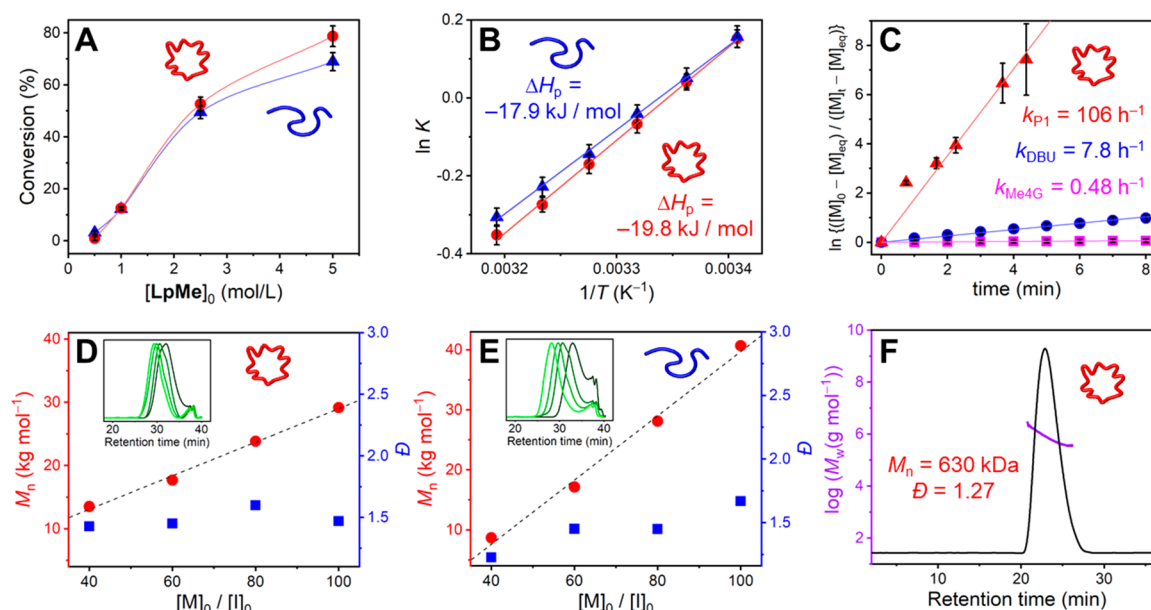


Figure 2. Controlled synthesis of cyclic and linear poly(LpMe)s by ROP. (A) Plot of conversion versus monomer concentration determined by ^1H NMR experiment (Figures S31 and S32). Conditions: $[\text{M}]_0:[\text{B}]_0:[\text{I}]_0 = 100:1:1$. (B) Van't Hoff analysis of polymer–monomer equilibrium using variable-temperature ^1H NMR experiment (Figures S33 and S34) and ring strains evaluated by the enthalpies of polymerization. Conditions: $[\text{M}]_0:[\text{B}]_0:[\text{I}]_0 = 50:1:1$, $[\text{M}]_0 = 2$ M, THF- d_8 . (C) Kinetics of ROP using P_1 , DBU, and $\text{Me}_4\text{guanidine}$ (Me_4G), and their pseudo-first-order rate constants determined by ^1H NMR (Figures S35–S37). Conditions: $[\text{M}]_0:[\text{B}]_0:[\text{I}]_0 = 1000:1:1$, $[\text{M}]_0 = 2.3$ M. M_n (red) and $\bar{D}$ (blue) versus monomer-to-initiator ratios for (D) cyclic and (E) linear poly(LpMe)s. Conditions: $[\text{M}]_0 = 4.5$ M. Insets: corresponding GPC analysis of crude polymers showed an increasing retention time for poly(LpMe)s made by using a larger $[\text{M}]_0:[\text{I}]_0$ ratios (black to green). (F) GPC analysis of the largest attempted cyclic poly(LpMe) after purification.

reached 80% at 5 M, which made mechanical agitation a challenge. We found 3–4.5 M is a good range where we observed good conversion (60–70%) producing viscous but flowable polymer solutions. The need for a high concentration is a signature of the low ring strain associated with LpMe, known to result mostly from torsional strain of the S–S bond.⁵³ Computationally, we predicted a ring strain of 29 kJ mol^{−1} (see Supporting Information S.13). Experimentally, we determined the polymerization enthalpy for LpMe is -20 kJ mol^{−1} for cyclic poly(LpMe) and -18 kJ mol^{−1} for linear polymer, both values comparable to Waymouth's determination for linear poly(LpMe)³⁰ of -15 kJ mol^{−1}. A ceiling temperature of ~ 300 K was then determined for both types of poly(LpMe)s, a useful trait for catalytic recycling of monomers without the need of high temperature (*vide infra*).

The polymerization kinetics for cyclic poly(LpMe) is highly dependent on the choice of base (Figure 2C). The observed rate enhancements correlate with the increasing size of the base/counter-cation, consistent with our expectation of ion-pairing in THF ($\epsilon_r = 7.6$).⁵⁴ The larger counter-cation leads to less ion-pairing and stronger nucleophilicity of propagating chain ends for a faster polymerization rate.⁵⁵ In particular, P_1 -initiated⁵⁶ polymerization reached >90% completion within 4 min and equilibrated in 10 min.

The ROP of LpMe showed good chain end fidelity and high livingness of polymerization (Figure 2D,E), as evident by the linear relationship between molecular weights (M_n) and M/I ratios. Both cyclic and linear poly(LpMe)s contain observable quantities of cyclic oligomers (insets, Figure 2D,E). While removable by precipitating into MeOH, the co-existence of these oligomers in the mixture resulted in higher M_n of the polymer than theoretically expected. We observed $\bar{D}$ of 1.4 at the equilibrium conversion of 70% (Table 1) and the value

increase to 1.5 at 80% conversion (Figure 2D,E). We expect that further improvement on $\bar{D}$ will require monomers with a higher ring strain⁵⁷ to increase the rate difference between propagation and chain transfer, which is ~ 600 for the current system.⁵⁸

The systematic studies on factors controlling the polymerization allowed us to synthesize high molecular weight poly(LpMe). We used an arbitrarily large M/I ratio of 50000:1 for this attempt. The high viscosity of the reaction mixture eventually led to a self-supported gel. The measured M_n of 630 kDa (Figure 2F) is most likely limited by formation of catenated cyclic polymers.²³ Overall, the good yield and reasonable $\bar{D}$ of 1.27 suggest the preparation of high poly(disulfide)s is practicable.

We next examined the reversibility of thiolate-disulfide exchange in performing catalytic depolymerization of poly(disulfide)s to recycle monomers (Figures 3A and S41). Both

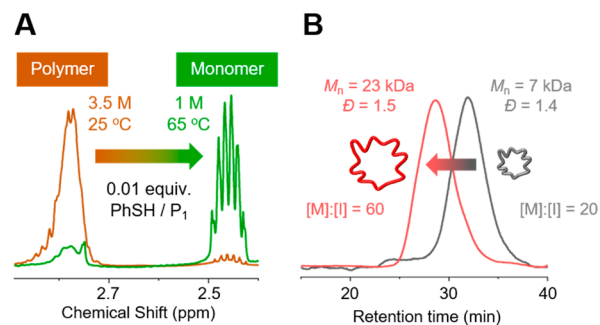


Figure 3. (A) Monomer regeneration from catalytic degradation of cyclic poly(LpMe) showed 95% purity. (B) GPC analysis showed ring expansion of cyclic poly(LpMe).

cyclic and linear poly(LpMe)s were tested. With dilution at 1 M and heating at 65 °C, the initiator became a catalyst for depolymerization (0.01 equiv). In one hour, 95% of cyclic poly(LpMe) with a 30 kDa molecular weight was recycled into monomers with PhSH/P₁, while 92% of linear polymer was recycled with BnSH/P₁. The use of catalytic amount of reagents and mild heating reflect robust recyclability of poly(disulfide)s.

To examine the reinitiation process,⁵⁹ we performed a ring expansion experiment. A purified small-molecular-weight polymer was subject to PhSH/P₁ reinitiation condition along with additional monomer (Figure 3B). Complete consumption of the initial polymer was observed. The GPC analysis confirmed both polymers to display cyclic geometry (Figure S42), suggesting the smaller rings have been expanded to larger ones.

Catalytic depolymerization and ring-expansion confirmed PhSH's ability to activate disulfide bonds in a polymer. The presence of nucleophilic chain ends is expected for a linear chain growth mechanism. Termination of these reactive groups by electrophilic quenchers is furthermore expected to trap the linear chain topology and thus preclude cyclic polymer formation. This behavior was not observed for PhSH initiated polymerization. Therefore, the high cyclic selectivity strongly suggests that the PhSH-initiated ROPs follow a ring-expansion mechanism,³⁵ in which rings are dominant over the propagating linear chains with the equilibrium position reflecting selectivity between the two topologies.

We used DFT computations to assess the influence of PhS[−] on the ring–chain equilibrium compared to alkyl thiolates (Figure 4). We evaluated nucleophilic attack on LpMe

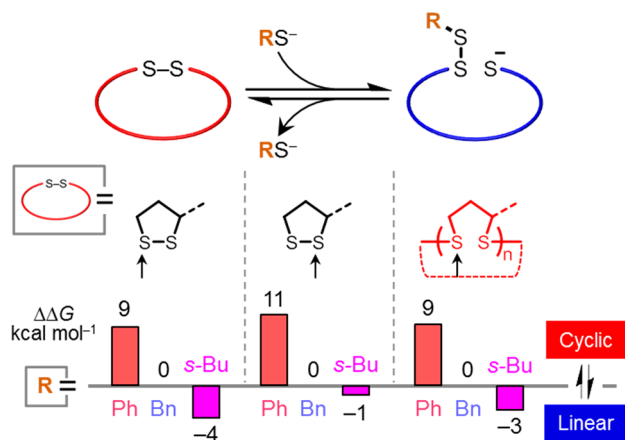


Figure 4. DFT analysis of ring-opening equilibrium for phenyl and alkyl thiolates. All computed free energies are presented as ring-opening reactions relative to BnSH in each series. Arrows indicate modeled nucleophilic attacks. *sec*-Butyl (*s*-Bu) thiolate represents a propagating chain end of poly(LpMe) (Figure S47).

monomer (initiation and propagation) and polymer mid-chain (chain transfer or backbiting) (see Supporting Information S.14). Across all three series, PhS[−] favors ring-closing reactions by 9–13 kcal mol^{−1} compared to alkyl thiolate, supporting our hypothesis that rings are favored over linear chains when nucleofugal PhS[−] is used as the initiator.

Finally, we applied PhSH/P₁ initiation on a wider range of lipoate monomers to include aromatic, polar, and fluoruous substituents (see Supporting Information S.15). All monomers were successfully polymerized to yield cyclic polymers with

good conversion (~70%) and consistent dispersity ($\bar{D} = 1.5$), demonstrating the generality of our methods.

In summary, we have developed a controlled synthesis of cyclic poly(disulfide)s by modulating the nucleophilicity of initiators. The cyclic poly(disulfide)s is achieved when arylthiol initiators are used. The excellent tunability of polymerization allows for preparation of large polymers (630 kDa) as well as catalytic depolymerization into monomers. Overall, the method represents a new paradigm in approaching the synthesis of cyclic polymers.

■ ASSOCIATED CONTENT

§ Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/jacs.9b08957.

Experimental procedures, NMR and mass spectra, GPC chromatograms, Mark–Houwink–Sakurada plots, logarithm of M_w vs retention time plots, and XYZ coordinates of optimized geometries, including Figures S1–S64 and Tables S1–S9 (PDF)

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Notes

The authors declare no competing financial interest.

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■ REFERENCES

- (1) Singh, R.; Whitesides, G. R. Thiol-disulfide Interchange. In *Supplement S: The Chemistry of Sulfur-Containing Functional Groups*; Patai, S., Rapport, Z., Eds.; Wiley, 1993.
- (2) Fass, D.; Thorpe, C. Chemistry and enzymology of disulfide cross-linking in proteins. *Chem. Rev.* **2018**, *118*, 1169–1198.
- (3) Schafer, F. Q.; Buettner, G. R. Redox environment of the cell as viewed through the redox state of the glutathione disulfide/glutathione couple. *Free Radical Biol. Med.* **2001**, *30*, 1191–1212.
- (4) Aoki, D.; Aibara, G.; Uchida, S.; Takata, T. A rational entry to cyclic polymers via selective cyclization by self-assembly and topology transformation of linear polymers. *J. Am. Chem. Soc.* **2017**, *139*, 6791–6794.
- (5) Son, S.; Namgung, R.; Kim, J.; Singha, K.; Kim, W. J. Bioreducible polymers for gene silencing and delivery. *Acc. Chem. Res.* **2012**, *45*, 1100–1112.
- (6) Andren, O. C. J.; Fernandes, A. P.; Malkoch, M. Heterogeneous rupturing dendrimers. *J. Am. Chem. Soc.* **2017**, *139*, 17660–17666.
- (7) Yu, B.; Zheng, Y.; Yuan, Z.; Li, S.; Zhu, H.; De La Cruz, L. K.; Zhang, J.; Ji, K.; Wang, S.; Wang, B. Toward direct protein S-

persulfidation: A prodrug approach that directly delivers hydrogen persulfide. *J. Am. Chem. Soc.* **2018**, *140*, 30–33.

(8) Hashim, P. K.; Okuro, K.; Sasaki, S.; Hoashi, Y.; Aida, T. Reductively cleavable nanocaplets for siRNA delivery by template-assisted oxidative polymerization. *J. Am. Chem. Soc.* **2015**, *137*, 15608–15611.

(9) Fu, J.; Yu, C.; Li, L.; Yao, S. Q. Intracellular delivery of functional proteins and native drugs by cell-penetrating poly(disulfide)s. *J. Am. Chem. Soc.* **2015**, *137*, 12153–12160.

(10) Qian, L.; Fu, J.; Yuan, P.; Du, S.; Huang, W.; Li, L.; Yao, S. Q. Intracellular delivery of native proteins facilitated by cell-penetrating poly(disulfide)s. *Angew. Chem., Int. Ed.* **2018**, *57*, 1532–1536.

(11) Chuard, N.; Gasparini, G.; Moreau, D.; Lorcher, S.; Palivan, C.; Meier, W.; Sakai, N.; Matile, S. Strain-Promoted Thiol-Mediated Cellular Uptake of Giant Substrates: Liposomes and Polymersomes. *Angew. Chem., Int. Ed.* **2017**, *56*, 2947–2950.

(12) Carnall, J. M.; Waudby, C. A.; Belenguer, A. M.; Stuart, M. C.; Peyralans, J. J.; Otto, S. Mechanosensitive self-replication driven by self-organization. *Science* **2010**, *327*, 1502–1506.

(13) Zhang, Q.; Deng, Y.; Luo, H.; Shi, C.; Geise, G. M.; Feringa, B. L.; Tian, H.; Qu, D. Assembling a natural small molecule into a supramolecular network with high structural order and dynamic functions. *J. Am. Chem. Soc.* **2019**, *141*, 12804–12814.

(14) Samuel, N. K. P.; Singh, M.; Yamaguchi, K.; Regen, S. L. Polymerized-depolymerized vesicles. Reversible thiol-disulfide-based phosphatidylcholine membranes. *J. Am. Chem. Soc.* **1985**, *107*, 42–47.

(15) Canadell, J.; Goossens, H.; Klumperman, B. Self-healing materials based on disulfide links. *Macromolecules* **2011**, *44*, 2536–2541.

(16) Yoon, J. A.; Kamada, J.; Koynov, K.; Mohin, J.; Nicolaÿ, R.; Zhang, Y.; Balazs, A. C.; Kowalewski, T.; Matyjaszewski, K. Self-healing polymer films based on thiol–disulfide exchange reactions and self-healing kinetics measured using atomic force microscopy. *Macromolecules* **2012**, *45*, 142–149.

(17) Kleine, T. S.; Nguyen, N. A.; Anderson, L. E.; Namnabat, S.; LaVilla, E. A.; Showghi, S. A.; Dirlam, P. T.; Arrington, C. B.; Manchester, M. S.; Schwiegerling, J.; Glass, R. S.; Char, K.; Norwood, R. A.; Mackay, M. E.; Pyun, J. High refractive index copolymers with improved thermomechanical properties via the inverse vulcanization of sulfur and 1,3,5-triisopropenylbenzene. *ACS Macro Lett.* **2016**, *5*, 1152–1156.

(18) Zhang, Y.; Glass, R. S.; Char, K.; Pyun, J. Recent advances in the polymerization of elemental sulphur, inverse vulcanization and methods to obtain functional chalcogenide hybrid inorganic/organic polymers (CHIPs). *Polym. Chem.* **2019**, *10*, 4078–4105.

(19) Fortman, D. J.; Snyder, R. L.; Sheppard, D. T.; Dichtel, W. R. Rapidly reprocessable cross-linked polyhydroxyurethanes based on disulfide exchange. *ACS Macro Lett.* **2018**, *7*, 1226–1231.

(20) Singh, P.; Rheinhardt, J. H.; Olson, J. Z.; Tarakeshwar, P.; Mujica, V.; Buttry, D. A. Electrochemical capture and release of carbon dioxide using a disulfide-thiocarbonate redox cycle. *J. Am. Chem. Soc.* **2017**, *139*, 1033–1036.

(21) Preefer, M. B.; Oschmann, B.; Hawker, C. J.; Seshadri, R.; Wudl, F. High sulfur content material with stable cycling in lithium-sulfur batteries. *Angew. Chem., Int. Ed.* **2017**, *56*, 15118–15122.

(22) Bang, E.-K.; Lista, M.; Sforazzini, G.; Sakai, N.; Matile, S. Poly(disulfide)s. *Chem. Sci.* **2012**, *3*, 1752–1763.

(23) Endo, K.; Shiroy, T.; Murata, N.; Kojima, G.; Yamanaka, T. Synthesis and Characterization of Poly(1,2-dithiane). *Macromolecules* **2004**, *37*, 3143–3150.

(24) Kisanuki, A.; Kimpara, Y.; Oikado, Y.; Kado, N.; Matsumoto, M.; Endo, K. Ring-opening polymerization of lipoic acid and characterization of the polymer. *J. Polym. Sci., Part A: Polym. Chem.* **2010**, *48*, 5247–5253.

(25) Lai, Y.; Kuang, X.; Zhu, P.; Huang, M.; Dong, X.; Wang, D. Colorless, transparent, robust, and fast scratch-self-healing elastomers via a phase-locked dynamic bonds design. *Adv. Mater.* **2018**, *30*, No. e1802556.

(26) Liang, Z. A.; Meng, Y. Z.; Li, L.; Du, X. S.; Hay, A. S. Novel synthesis of macrocyclic aromatic disulfide oligomers by cyclo-depolymerization of aromatic disulfide polymers. *Macromolecules* **2004**, *37*, 5837–5840.

(27) Rosenthal, E. Q.; Puskas, J. E.; Wesdemiotis, C. Green polymer chemistry: living dithiol polymerization via cyclic intermediates. *Biomacromolecules* **2012**, *13*, 154–164.

(28) Bang, E. K.; Gasparini, G.; Molinard, G.; Roux, A.; Sakai, N.; Matile, S. Substrate-initiated synthesis of cell-penetrating poly-(disulfide)s. *J. Am. Chem. Soc.* **2013**, *135*, 2088–2091.

(29) Gasparini, G.; Bang, E. K.; Molinard, G.; Tulumello, D. V.; Ward, S.; Kelley, S. O.; Roux, A.; Sakai, N.; Matile, S. Cellular uptake of substrate-initiated cell-penetrating poly(disulfide)s. *J. Am. Chem. Soc.* **2014**, *136*, 6069–6074.

(30) Zhang, X.; Waymouth, R. M. 1,2-Dithiolane-derived dynamic, covalent materials: Cooperative self-assembly and reversible cross-linking. *J. Am. Chem. Soc.* **2017**, *139*, 3822–3833.

(31) Laurent, B. A.; Grayson, S. M. Synthetic approaches for the preparation of cyclic polymers. *Chem. Soc. Rev.* **2009**, *38*, 2202–2213.

(32) Yamamoto, T.; Tezuka, Y. Topological polymer chemistry: A cyclic approach toward novel polymer properties and functions. *Polym. Chem.* **2011**, *2*, 1930–1941.

(33) Bielawski, C. W.; Benitez, D.; Grubbs, R. H. An “endless” route to cyclic polymers. *Science* **2002**, *297*, 2041–2044.

(34) Xia, Y.; Boydston, A. J.; Yao, Y.; Kornfield, J. A.; Gorodetskaya, I. A.; Spiess, H. W.; Grubbs, R. H. Ring-expansion metathesis polymerization: Catalyst-dependent polymerization profiles. *J. Am. Chem. Soc.* **2009**, *131*, 2670–2677.

(35) Kaitz, J. A.; Diesendruck, C. E.; Moore, J. S. End group characterization of poly(phthalaldehyde): Surprising discovery of a reversible, cationic macrocyclization mechanism. *J. Am. Chem. Soc.* **2013**, *135*, 12755–12761.

(36) Roland, C. D.; Li, H.; Abboud, K. A.; Wagener, K. B.; Veige, A. S. Cyclic polymers from alkynes. *Nat. Chem.* **2016**, *8*, 791–796.

(37) Narumi, A.; Yamada, M.; Unno, Y.; Kumaki, J.; Binder, W. H.; Enomoto, K.; Kikuchi, M.; Kawaguchi, S. Evaluation of ring expansion-controlled radical polymerization system by AFM observation. *ACS Macro Lett.* **2019**, *8*, 634–638.

(38) Culkun, D. A.; Jeong, W.; Cshony, S.; Gomez, E. D.; Balsara, N. P.; Hedrick, J. L.; Waymouth, R. M. Zwitterionic polymerization of lactide to cyclic poly(lactide) by using N-heterocyclic carbene organocatalysts. *Angew. Chem., Int. Ed.* **2007**, *46*, 2627–2630.

(39) Guo, L.; Zhang, D. Cyclic poly(alpha-peptoid)s and their block copolymers from N-heterocyclic carbene-mediated ring-opening polymerizations of N-substituted N-carboxylanhydrides. *J. Am. Chem. Soc.* **2009**, *131*, 18072–18074.

(40) Guo, L.; Lahasky, S. H.; Ghale, K.; Zhang, D. N-Heterocyclic carbene-mediated zwitterionic polymerization of N-substituted N-carboxylanhydrides toward poly(alpha-peptoid)s: kinetic, mechanism, and architectural control. *J. Am. Chem. Soc.* **2012**, *134*, 9163–9171.

(41) Brown, H. A.; De Crisci, A. G.; Hedrick, J. L.; Waymouth, R. M. Amidine-mediated zwitterionic polymerization of lactide. *ACS Macro Lett.* **2012**, *1*, 1113–1115.

(42) Hong, M.; Chen, E. Y. Completely recyclable biopolymers with linear and cyclic topologies via ring-opening polymerization of gamma-butyrolactone. *Nat. Chem.* **2016**, *8*, 42–49.

(43) Cortez, M. A.; Godbey, W. T.; Fang, Y.; Payne, M. E.; Cafferty, B. J.; Kosakowska, K. A.; Grayson, S. M. The synthesis of cyclic poly(ethylene imine) and exact linear analogues: An evaluation of gene delivery comparing polymer architectures. *J. Am. Chem. Soc.* **2015**, *137*, 6541–6549.

(44) Yamamoto, T.; Yagyu, S.; Tezuka, Y. Light- and heat-triggered reversible linear-cyclic topological conversion of telechelic polymers with anthryl end groups. *J. Am. Chem. Soc.* **2016**, *138*, 3904–3911.

(45) Saxon, D. J.; Nasiri, M.; Mandal, M.; Maduskar, S.; Dauenhauer, P. J.; Cramer, C. J.; LaPointe, A. M.; Reineke, T. M. Architectural control of isosorbide-based polyethers via ring-opening polymerization. *J. Am. Chem. Soc.* **2019**, *141*, 5107–5111.

- (46) Ogawa, T.; Nakazono, K.; Aoki, D.; Uchida, S.; Takata, T. Effective approach to cyclic polymer from linear polymer: Synthesis and transformation of macromolecular [1]rotaxane. *ACS Macro Lett.* **2015**, *4*, 343–347.
- (47) Aoki, D.; Aibara, G.; Uchida, S.; Takata, T. A rational entry to cyclic polymers via selective cyclization by self-assembly and topology transformation of linear polymers. *J. Am. Chem. Soc.* **2017**, *139*, 6791–6794.
- (48) Bannin, T. J.; Kiesewetter, M. K. Poly(thioester) by organocatalytic ring-opening polymerization. *Macromolecules* **2015**, *48*, 5481–5486.
- (49) Garrido, G.; Koort, E.; Rafols, C.; Bosch, E.; Rodima, T.; Leito, I.; Roses, M. Acid-base equilibria in nonpolar media. Absolute pK(a) scale of bases in tetrahydrofuran. *J. Org. Chem.* **2006**, *71*, 9062–9067.
- (50) Zhu, J. B.; Watson, E. M.; Tang, J.; Chen, E. Y. A synthetic polymer system with repeatable chemical recyclability. *Science* **2018**, *360*, 398–403.
- (51) Laurent, Q.; Sakai, N.; Matile, S. The opening of 1,2-dithiolanes and 1,2-diselenolanes: Regioselectivity, rearrangements, and consequences for poly(disulfide)s, cellular uptake and pyruvate dehydrogenase complexes. *Helv. Chim. Acta* **2019**, *102*, No. e1800209.
- (52) Dubois, P.; Coulembier, O.; Raquez, J.-M. *Handbook of ring-opening polymerization*; Wiley-VCH: Weinheim, 2009.
- (53) Burns, J. A.; Whitesides, G. M. Predicting the stability of cyclic disulfides by molecular modeling: Effective concentrations in thiol-disulfide interchange and the design of strongly reducing dithiols. *J. Am. Chem. Soc.* **1990**, *112*, 6296–6303.
- (54) Liu, Y.; Sengupta, A.; Raghavachari, K.; Flood, A. H. Anion binding in solution: Beyond the electrostatic regime. *Chem.* **2017**, *3*, 411–427.
- (55) Chae, C.-G.; Bak, I.-G.; Lee, J.-S. Propagation-inspired initiation of an aliphatic sodium amidate for the living anionic homo- and copolymerization of isocyanates: Access to the multi-blocky sequence distribution of binary comonomers. *Macromolecules* **2018**, *51*, 10083–10094.
- (56) Zhang, L.; Nederberg, F.; Pratt, R. C.; Waymouth, R. M.; Hedrick, J. L.; Wade, C. G. Phosphazene bases: A new category of organocatalysts for the living ring-opening polymerization of cyclic esters. *Macromolecules* **2007**, *40*, 4154–4158.
- (57) Zong, L.; Bartolami, E.; Abegg, D.; Adibekian, A.; Sakai, N.; Matile, S. Epidithiodiketopiperazines: strain-promoted thiol-mediated cellular uptake at the highest tension. *ACS Cent. Sci.* **2017**, *3*, 449–453.
- (58) Singh, R.; Whitesides, G. M. Degenerate intermolecular thiolate-disulfide interchange involving cyclic five-membered disulfides is faster by $\sim 10^3$ than that involving six- or seven-membered disulfides. *J. Am. Chem. Soc.* **1990**, *112*, 6304–6309.
- (59) Sakai, N.; Lista, M.; Kel, O.; Sakurai, S.; Emery, D.; Mareda, J.; Vauthey, E.; Matile, S. Copolymerization of lipoic acid with 1,2-dithiane and characterization of the copolymer as an interlocked cyclic polymer. *J. Am. Chem. Soc.* **2011**, *133*, 15224–15227.